



Joshua Olusoji (Attempt 1 of 1)

Test Information

QUESTION 1: CALCULATED NUMERIC

5

out of 5 points

Consider a process described by the following transfer function

$$G_P(s) = \frac{(-s+1)}{(10s+1)(2s+1)}$$



We wish to approximate this process with an FOPTD. Calculate the Delay time of the approximate model.

Given Answer: 2

Correct Answer: 2

Answer range +/- 0 (2 - 2)

QUESTION 2: CALCULATED NUMERIC

5

out of 5 points

Determine the Open Stem Travel (%Valve Open) of a 4in inline control valve in an 8in NPS Schedule 40 pipeline, with inherent characteristic given in **Table 1** in the **CHE Calculator** that is used to regulate the flow of liquid propane. The service conditions are: Flowing temperature T=50 °F, volumetric design flow rate Q=250 gpm, upstream pressure P₁=350 psia and downstream pressure P₂=300 psia. Express your result in percent travel (for instance if you



answer is 25.3% enter the value 25). Assume that the **Peng-Robinson** Equation of State can be used for the calculation of the thermal properties of propane and the **Rackett** equation for the liquid density. **Round your answer to a whole number.**

Given Answer: 45

Correct Answer: 45

Answer range +/- 0 (45 - 45)

Consider the mixing of two streams: **Stream 1** with flow rate $F_1=5$ kg/s and temperature $T_1=25$ °C and **Stream 2** with flow rate $F_2=5$ kg/s and temperature $T_2=75$ °C. The streams are mixed in a tank with constant mass (from a perfect level controller) equal to $M=100$ kg. We wish to design a controller that would regulate the temperature of the effluent stream from the tank by manipulating the flow rate of **Stream 1**. Derive the process transfer function for this control loop. Your answer must contain only numerical values and the Laplace transformation symbol s . Temperature units in the final answer must be in °C. Assume that the heat capacities of all streams and that of the mass in the mixing tank are all equal.

Given
Answer: $\frac{-2.5}{10s+1}$

Correct
Answer: [None]



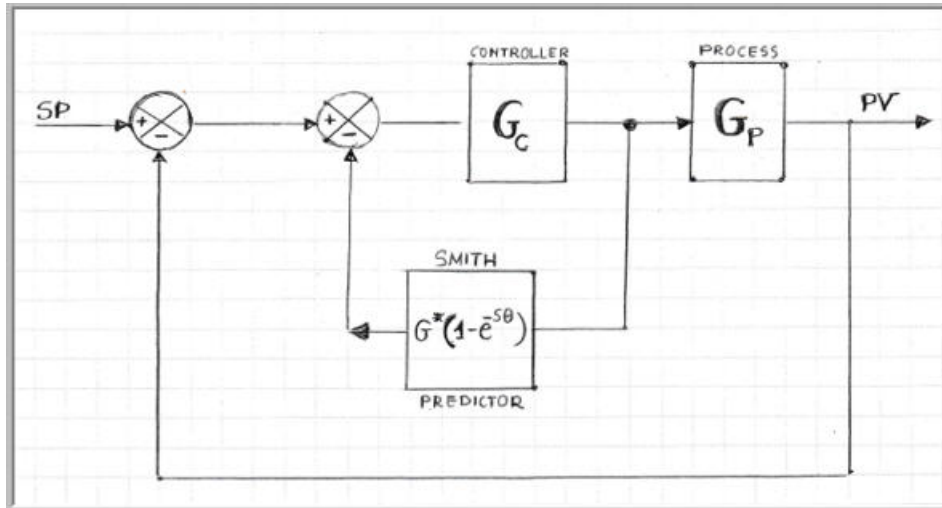
Response
Feedback:

For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

B	<i>I</i>	<u>U</u>	S	Paragraph	Open Sans,ari...	10pt	≡
≡	A		<i>I_x</i>				
$\times^2$	$\times_2$						
		...					

P 0 WORDS POWERED BY TINY

Time delays frequently occur in the process industries because of the presence of transportation lags, recycle loops and sensor inertia. Time delays result in reduction of the controller gain in order to maintain stability and as a result make the controller performance sluggish. To improve the performance of loops with time delays, advanced control strategies have been developed over the years. One such strategy is known as the **Smith predictor** and it consists of an ordinary feedback loop plus an inner loop that introduces an extra term directly into the feedback path. A control block diagram of the Smith predictor controller is given below:



where $G_P(s) = G_P^*(s) e^{-s\theta}$ and $G_P^*(s)$ is the process transfer function free of the time delay. Consider the



following first order process $G_P(s) = \frac{1.3e^{-3.7s}}{13.7s+1}$. This chemical process is controlled with a PI feedback

controller **with the Smith-Predictor in the loop**. The delay time in the Smith-Predictor is set at $\theta = 3.7$. Perform a simulation for a unit step change in the set point and calculate the **overshoot** and the **settling time**. Enter your numerical results separated by comma in the textbox below:

Given 0.498, 47.8

Answer:

Correct [None]

Answer:

Response For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

Feedback:

B	I	U	S	Paragraph	Open Sans, ari...	10pt		

P

0 WORDS POWERED BY TINY

QUESTION 5: CALCULATED NUMERIC

5

out of 5 points

A controller sends an 12mA analog signal to an I/P converter. The pressure signal from the I/P converter is sent to the control valve. What is the valve opening as a percentage of the total travel, assuming total travel corresponds to 100% (fully open valve)



Given Answer: 50

Correct Answer: 50

Answer range +/- 0 (50 - 50)

QUESTION 6: MULTIPLE ANSWER

10

out of 10 points

Consider the process transfer function: $G_P(s) = \frac{1}{-5s+1}$ Assume that the controller gain K_c is positive. Using the Routh–Hurwitz criterion, determine which the following controllers can stabilize the loop. Select all that apply.



Given ☒ Proportional Derivative (PD): $G_c(s) = K_c(1 + \tau_D s)$

Answers:



☒ Proportional Integral Derivative (PID): $G_c(s) = K_c\left(1 + \frac{1}{\tau_I s} + \tau_D s\right)$

Correct ☒ Proportional Derivative (PD): $G_c(s) = K_c(1 + \tau_D s)$

Answers:



☒ Proportional Integral Derivative (PID): $G_c(s) = K_c\left(1 + \frac{1}{\tau_I s} + \tau_D s\right)$

QUESTION 7: CALCULATED NUMERIC

5

out of 5 points

Using the Bode stability analysis determine the minimum value of the reset time τ_I for stability of a **PI** controller with gain $K_C = 0.3$ applied to a process with transfer function

$$G_P(s) = \frac{e^{-0.5s}}{s+1}$$



Round your numerical answer into **four decimal digits**.

Given Answer: 0.1118

Correct Answer: 0.1118

Answer range +/- 0.0001 (0.11170 - 0.11190)

out of 5 points

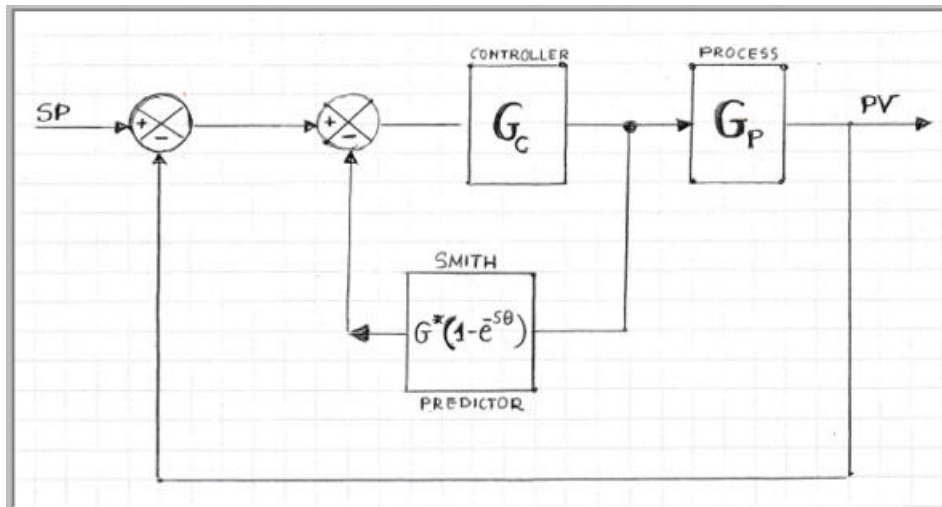
$$G_P(s) = \frac{0.5e^{-5s}}{(20s+1)(5s+1)}$$


Answer range +/- 0.0001 (11.95970 - 11.95990)

out of 10 points

[illegible]

Time delays frequently occur in the process industries because of the presence of transportation lags, recycle loops and sensor inertia. Time delays result in reduction of the controller gain in order to maintain stability and as a result make the controller performance sluggish. To improve the performance of loops with time delays, advanced control strategies have been developed over the years. One such strategy is known as the **Smith predictor** and it consists of an ordinary feedback loop plus an inner loop that introduces an extra term directly into the feedback path. A control block diagram of the Smith predictor controller is given below:



where $G_P(s) = G_P^*(s) e^{-s\theta}$ and $G_P^*(s)$ is the process transfer function free of the time delay. Derive the



closed-loop transfer function $G_{CL}(s) = \frac{PV}{SP}$ and enter your final formula in the textbox below:

Given

Answer:

$$\frac{G_c(s) G_p(s)}{1 + G_c(s) G_p^*(s)}$$

Correct

[None]

Answer:

Response
Feedback:

For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

B	I	U	S	Paragraph	Open Sans, ari...	10pt	⋮
⋮	A						
$\times^2$	$\times_2$						

P 0 WORDS POWERED BY TINY

A PD controller would work to stabilize the closed-loop response. It would have the form:

$$G_c(s) = K_c(1 + \tau_D s) \text{ with stability conditions: } K_c > 0 \text{ and } K_c \tau_D > 5$$

[None]

Answer:

For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

[illegible]